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(1) Research Program/Project/Study Title: Sorghum as Vehicle for Rural Development in the Province of Capiz	
Proponent Name:	Freddie L. Baranda
Academic Rank:	Instructor I
Co-Authors (with academic rank):	Dr. Ryan T. Sarimong – Asso Prof 5 Roger L. Berganio - Asst Prof 4 Francisco Lutao, Jr. – Inst 1
Campus/Satellite College/Address:	Burias
Telephone/Cell Phone/Fax Number:	
E-mail Address:	

Program or Project has component (if no proceed to 2)? ☒ Yes ☐ No

Component Title 1:	Characterization of the Bio-Chemical and Agro-Ecological Attributes of Typical Sorghum Farms in Capiz
Component Leader:	Freddie L. Baranda, Ryan T. Sarimong, Roger L. Berganio
Academic Rank:	Inst 1 and Asst Prof 3
Component Title 2:	Varietal Collection, Identification, Seed Dispersal and Characterization of Sorghum Varieties in Capiz
Component Leader:	Freddie L. Baranda and Roger L. Berganio
Academic Rank:	
Component Title 3:	Adaptability Trials, Pilot Testing and Production Performance of the Climate Resilient Sorghum Varieties Grown in Multi-locations in the selected municipalities in the Province of Capiz
Component Leader:	Roger L. Berganio and Freddie L. Baranda, Alberto Valdemar
Academic Rank:	CAF Faculty, MA
Component Title 4:	Bio-physico and chemical Characteristics of Sorghum Cultivars Influenced by the Levels of Fertilizers
Component Leader:	Ryan T. Sarimong, Freddie L. Baranda and Roger L. Berganio,
Academic Rank:	
Component Title 5:	Characterization of Grains, Milling Performance, and Nutrient Content Analyses of Different Sorghum Cultivars
Component Leader:	Roger L. Berganio and Freddie L. Baranda
Academic Rank:	
Component Title 6:	Proximate Analysis and Nutrient Composition of Pelletized Urea-Treated Sorghum Stalks
Component Leader:	John King N. Layos, Roger L. Berganio and Freddie L. Baranda
Component Title 7:	Impact, Production and Status of Sorghum Industry in the Province of Capiz
Component Leader:	Freddie L. Baranda, Roger L. Berganio and Francisco Lutao Jr., Connie Labrador, Alberto Valdemar, Elsie Baranda
	Faculty CAF

(2) Category (please check) ☒ Social Science/Education /Humanities & other related fields ☒ Natural/Biological Sciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center
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	☐ FFRDC Fisheries and Food Research Development Center ☒ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	Mambusao, Jamindan, Tapaz
Province:	Capiz
Region:	VI

(5) Classification Research: Development: ☒ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
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(7) Sector/Commodity Sorghum	(8) Discipline/Field of Specialization: Agriculture, and management
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(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

Sorghum (*Sorghum bicolor* L. Moench) is widely used as food, feed, fiber, and bioenergy crop. The grain is used as food or feed; the stem can be used as a source of fiber, fuel, and as feedstock (Yi-Hong Wang et.al, 2016). It also used as ingredient in snack foods and baking and brewing products, and even as a meat extender (<https://www.purdue.edu › 0004.Hamaker.sorghum.htm>)

Sorghum is one of the identified sources of biomass which potential to generate bio-ethanol to fuel gasoline engine is well appreciated. The juice from the stalk can be processed in the form of syrup which can be later be used as sweeteners for pancakes, cornmeal mush, and other hot cereals. It can be a source of sugar, alcohol, jaggery and animal feeds (Gregorio and Launio, 2007).

Sorghum is a nutrient-packed grain which is rich in vitamins and minerals like B vitamins, magnesium, potassium, phosphorus, iron, and zinc. It's also an excellent source antioxidants such as flavonoids, phenolic acids, and tannins, and protein. (<https://www.healthline.com/nutrition/sorghum#bottom-line>)

Sorghum has a proximate composition, amino acid contents, and nutritional value similar to those of maize. However, due to its lower fat content, sorghum usually has a slightly lower gross, digestible, and metabolizable energy.

However, due to rapid population growth, our produce particularly rice is enough only for the 90 percent population in the Philippines (PSA 2018) hence sorghum has a potentials which could be utilized in lieu of rice and corn as staple food for the Filipino. But one of the major problems in terms of sorghum particularly in Capiz are limited number of farmers in growing sorghum crop hence R&D is imperative to assess its food chain status, its production potential, market, adaptability and suitability, among others hence this study was conceptualized.

Sorghum is an excellent feed for livestock and companion animals. The feeding value of sorghum for livestock species is generally considered 95% or more of the feeding value of yellow dent maize. The brown sorghums are considered to have 80–85% the feeding value of maize due to the presence of condensed tannins that reduced digestibility. Regardless of type, sorghum must be properly processed to enhance its digestibility. Popping, steam flaking, and reconstitution are used to prepare sorghum grain in beef cattle feedlots. Grinding, rolling, crushing, and pelleting are used for poultry and swine feeds.

However, despite of numerous benefits that sorghum plants can offers, however few farmers in Capiz adapt and raise sorghum crop as their staple food as substitute for rice and corn or even as use as feed ingredient for their farms animal due to strangeness of the said crop therefore massive promotion campaign is therefore imperative.

Although there are existing farmers sorghum in the province of Capiz, however there is no enough data to support in terms of production status of sorghum industry in the Province of Capiz, its bio-chemical and agro-ecological attributes of typical sorghum farms, suitable sorghum varieties, record on the quality of sorghum and its characteristics when grown under Capiz condition, and also its contribution to food production industry in Capiz. With these premise this study will be conducted.

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)

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(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

Component 1: Characterization of the Bio-Chemical and Agro-Ecological Attributes of Typical Sorghum Farms in the Panay Island

1. Describe the population density, vegetation, topographic and major farm resources of the typical sorghum farms in the study area;
2. Assess the soil fertility status and water resources of the sorghum farms;
3. Describe the cultural practices employed the sorghum farmers and its relation to sorghum farming;
4. Describe the environmental problems associated with sorghum farming.

Component 2: Varietal Collection, Identification, Seed Dispersal and Characterization of Sorghum Varieties in Capiz

1. To undertake collection the different sorghum varieties in the province of Capiz;
2. To identify different sorghum cultivars in the locality;
3. To characterize the different sorghum varieties in terms of growth, yield, maturity, grain characteristics and milling characteristics;
4. To create database for sorghum varieties as basis for commercial and industrial and economic use;
5. To use these sorghum varieties as materials for seed dispersal and for further experimentation

Component 3: Adaptability Trials and Pilot Testing and Production Performance of the Climate Resilient Sorghum Varieties Grown in Multi-locations in the selected municipalities in the Province of Capiz

1. To determine which sorghum varieties can thrive under Capiz Agro-climatic Conditions;
2. To assess the growth and yield performance and grain characteristics of the different sorghum varieties;
3. To assess the economics of production of sorghum

Component 4: Bio-physico-chemical characteristics of sorghum cultivars influenced by the the levels of fertilizers

1. Determine the physico-chemical characteristics of different sorghum cultivars ; and
2. Find out which level of applied fertilizers will influenced the physico-chemical characteristics of Sorghum.

Component 5: Characterization of grains, Milling Performance, and Nutrient Content Analysis of Sorghum

1. Evaluate the characteristics sorghum cultivars in term of purity, moisture content, and degree of insect infestation;
2. Determined the dehulling recovery and potential milling;
3. Measure the physical dimensions of milled grains sorghum in terms of head grains length, width, and thickness;
4. Assess the quality of milled sorghum in terms of percent head grains, broken, brewers, and other milling components
5. To assess the qualitative rating of milled sorghum based on degree of milling
6. Analyze the nutrient content of milled grains sorghum

Component 6. Proximate analysis and nutrient composition of different chemically-treated pelletized sorghum stalks

1. To determine the nutrient composition of pelletized sorghum stalks subjected to various chemical treatments; Urea, Lime, Sodium hydroxide and Potassium hydroxide

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2. To evaluate the physico-chemical analysis and nutrient composition of chemically treated pelletized sorghum stalks specifically in terms of: Dry Matter (DM), Organic Matter (OM), Crude Fiber (CF), Ash, Crude Protein (CP) and Neutral Detergent Fiber (NDF)

Component 7. Impact, Production and Status of Sorghum Industry in the Province of Capiz

1. To describe the production sorghum per municipalities in the province of Capiz
2. To describe the status of sorghum industry in the province of Capiz
3. To assess the income of sorghum farmers in capiz

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of –the-art of current technology / information from which the project proposal will take-off).

Sorghum (*Sorghum bicolor* (L.) Moench) is one of the most important cereal crops widely grown for food, feed, fodder, forage and fuel in the semi-arid tropics of Asia, Africa, the Americas and Australia. In spite of rapid decreases in the area of sorghum in Asia, the production level has been maintained owing to the adoption of high yielding hybrids. Though impressive progress has been made in improving the sorghum cultivars for resistance to biotic and abiotic challenges, grain mold, shoot fly and terminal drought haunt sorghum growers across Asia. The International Crops Research Institute for the Semi-Arid Tropics (ICRISAT) and national programs in India, China and Thailand are working on genetic enhancement of sorghum for high yield, grain mold and shortly resistance. In addition, the trait focus at ICRISAT includes post rainy season adaptation encompassing terminal drought tolerance, genetic enhancement for high Fe and Zn contents in grain and sweet stalk traits for ethanol and animal feed production. Genetic and cytoplasmic diversification of hybrid parents for key traits is critical for sustaining the productivity across different production systems. The grain and stover quality need special attention in sorghum improvement research to enhance the market value of sorghum. A brief description is provided of the progress made at ICRISAT in partnership with national programs in recent years in these areas of sorghum genetic enhancement. The sorghum area in Asia decreased from 23 million ha to 11 million ha between the early 1970s and 2007. However, production increased from 19 million t in the early 1970s to 21 million t in the late 1970s, but decreased thereafter to 11 million t in 2006. Yield has increased from 800 kg ha⁻¹ in the early 1970s to 1,000 kg ha⁻¹ in 2006 (Belum et al., 2010).

“Sorghum: nutrients, bioactive compounds, and the potential impact on human health,” the authors talk about the composition of this cereal and especially the virtues of these on human health. We know that sorghum is mainly composed of starch that is digested more slowly than other cereals that are low digestibility, protein and unsaturated fat and is a source of some vitamins and minerals. In addition, most sorghum varieties are rich in phenolic compounds, especially 3 – deoxyanthocyanidins and tannins. The results obtained in vitro and in animals have shown that phenolic compounds and the soluble compounds (polycosanols) highlighted in sorghum balance or stabilize the intestinal microbiota and the parameters associated with obesity, oxidative stress, inflammation, diabetes, dyslipidemia, hypertension and cancer. In conclusion, sorghum would be a source of nutrients and bioactive compounds, especially 3 - deoxyanthocyanidins, tannins and polycosanols, which advantageously modulate, in vitro (Morais et al., 2015)

In the study of conducted by Launio 2017 which was conducted at the CapSU [Capiz State University], Mambusao, Capiz in 2007 which aims to evaluate the growth and yield performance of ratooned ICRISAT Sweet Sorghum varieties under local conditions, response to various population densities, and NPK fertilizers, and to compare the performance of the ratooned to the main crop. Two studies were conducted in Split-plot design with three replications. The varieties studied were SPV 422, ICSV 93046, ICSV 700, ICSR 93034 and NTJ 2 while population densities included 100,000, 160,000, and 220,000/ha and levels of NPK fertilizer included the Control (No fertilizer), 80-30-30, and 160-60-60 kg NPK/ha. The growth parameters determined in this study for ratoon crop generally differed among the five varieties of sweet sorghum tested. NTJ2 and ICSR 93034 which matured 102 days after the harvest of the main crop appeared to be the earliest variety to mature. In both studies, ICSV 93046 and ICSV 700 produced the tallest plants and NTJ2 the shortest. In variety × density study, the

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highest panicle yield harvested was obtained by ICSR 93034 and NTJ2; but the plant population treatments did not influence this parameter; the SPV 422 produced the highest stripped stalk yield with 16.0 t/ha. However, the plant population had no effect on stripped stalk yield. On variety × fertilizer study, ICSV 93046 produced the highest stripped stalk yield. On variety x fertilizer study, ICSV 93046 produced the highest stripped stalk with 16.17 t/ha; the fertilized plants produced the heaviest stripped stalk yield. The performance of the ratoon crop clearly demonstrated the variability or suitability of sweet sorghum crop to be grown as ratoon. The main crop, however, generally performed better than the ratoon crop.

Nutrition

Sorghum has high antioxidant levels compared to other grains and vegetables and contains an impressive mix of phytochemicals. These phytochemicals may help control cholesterol levels, and sorghum's high phenolic content and high antioxidant properties have been shown to impact advanced glycation end products, which have been implicated as a complication of diabetes. In another limited study sorghum was found to elicit a low glucose response in healthy individuals suggesting that sorghum may be helpful in managing glucose and insulin levels. And remember, if you have celiac disease, sorghum is one of many great gluten-free grain options. Each whole grain offers a different mix of nutrients. Sorghum provides a good source (providing at least 10% of the recommended daily value) of fiber, magnesium, phosphorus, and vitamin B3 and is an excellent source (providing at least 20% of the recommended daily value) of manganese. <https://wholegrainscouncil.org/whole-grains-101/grain-month-calendar/sorghum-june-grain-month>

(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.)

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

Component 1: Characterization of the Bio-Chemical and Agro-Ecological Attributes of Typical Sorghum Farms in Capiz

The study will be conducted in sorghum farms in the selected barangays in various municipalities in the province Capiz upon knowing the records from the DA as to number sorghum farmers in Capiz. Sampling method will be used to determine the appropriate sample size. Then the typical sorghum will be objectively sampled for bio-chemical and agro-ecological attributes to represent a sorghum farms in the area. Assessment of such major aspects of sorghum farms as topographic features or landscapes, vegetation, crop grown with sorghum plant, soil

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fertility status, water resources, and environmental problems among others will be done. Soil samples will be likewise collected and brought to the soil laboratory for chemical analysis. Some interview will be also done to farmers to secure and validate some pertinent information relative to the data gathered from their farms.

Component 2: Varietal Collection, Identification, Seed Dispersal and Characterization of Sorghum Varieties in Capiz

The study will be conducted in the selected municipality in the province of Capiz. A preliminary survey and collection of secondary information relative to the need of study will be done. This will be followed by an intensive and exhaustive collection of these sorghum varieties will be done in the identified municipalities. Samples of these sorghum varieties will be asked from the sorghum farmers and placed in suitable containers. They will be brought to the laboratory for characterization activities. It will be made sure that enough seeds will be secured in preparation for growing them in the field for further observation. After a series of trials and when is enough seed for each varieties seed dispersal will be done for the identified sorghum farmer adoptors. To facilitate and to enhance the characterization process, the collections will be categorized by color, types or groups among others, the grain characteristics will be assessed as follows: seed weight, seed length, seed width, surface texture, shape, husk color, among others.

Component 3. Adaptability Trials and Pilot Testing and Production Performance of the Climate Resilient Sorghum Varieties Grown in Multi-locations in the selected municipalities in the Province of Capiz

The study will be conducted in the different location in the province of Capiz. The study will be conducted with different varieties arranged RCB. Data in terms of growth and yield characteristics will be collected and will be analyzed using ANOVA and LSD test at 5% level of significance.

Component 4. Bio-physico and chemical Characteristics of Sorghum Cultivars Influenced by the Level of Fertilizers

The study will be conducted using the Split plot design with Sorghum Cultivars as mainplot and levels of fertilizers as the sub-plot. Factor A constituted with four varieties namely: V1, V2, V3 and V4 while Factor B Constituted with four treatment such as F1, F2, F3 and F4. Various parameters will be gathered in relevant to the bio-physico and chemical attributes of the sorghum cultivars (e.g. plant height, biomass, brix or sugar reading etc).

Component 5 Characterization of grains, milling performance, and nutrient content analysis of sorghum cultivars

Sorghum grains that will be used in this study will be taken from the harvested sorghum from component 2 study. Samples will be sundried to 14% MC and will be stored for about 3 months, then, will be brought to laboratory seeds of NFA for analysis. Each variety of sorghum will be having a sample of three 250-gram to be submitted for laboratory evaluation. The following are the parameters to be determined in the laboratory analyses: Moisture Content & purity, Degree of Infestation, Dehauling recovery, Potential milling Recovery (PMR), Milled grains grading, Milling degree and Physical attributes. For this parameter 10 pieces head grains will be randomly selected per variety for the measurement of length, width and thickness of the grains. This will be done using the micrometer caliper from tip to tip of the kernel. The width of kernel will be measured from the widest part of the head grains while the thickness will be measured from the thickest portion of the grain (kernel). In terms of Nutrient Content Analyses (milled sorghum grains) will be sent to laboratory in the region capable of analyzing the nutrient content of the sorghum grains

Component 6. Methodology:

Sorghum stalks will be collected after harvest and shredded before the application of various chemical treatments and subsequently pelletized. After the chemical treatment and pelletizing process is done, the conventional methods to be used for chemical analysis of the different chemically treated pelletized sorghum stalks will be analyzed using the methods of Association of Official Analytical Chemists (AOAC 2005).

Dietary Treatments and Experimental Design

The chemical analysis of the following treatments will be analyzed through an *in situ* digestibility trial consisting of five test diets with three (3) replicates and placed inside in a single uniform rumen environment of a fistulated Brahman bull. The chemically treated test diets to be tested will be the following: T₀ – Pelletized sorghum stalks (Control), T₁ – Pelletized Urea-treated sorghum stalks, T₂ – Pelletized Lime-treated sorghum stalks, T₃ – Pelletized Sodium hydroxide sorghum stalks, T₄ – Pelletized Potassium hydroxide sorghum stalks. The parameters to be gathered are the following: %Dry Matter Digestibility (%DMD), %Crude Protein Digestibility c(%DMD), %Neutral Detergent Fiber Digestibility (%NDFD), and %Organic Matter Digestibility (%OMD).

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Component 7.

The primary data will be collected using interview schedule while the secondary data will be collected from the concern agencies. The data will be consolidated using MS excel. The data results will be presented in Frequency Table, percentage will be computed.

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart.

Research Objectives	Year 1						Year 2			
	July	Aug	Sept	Oct	Nov	Dec	Jan	Feb	Mar	Apr
1. Field work/observation										
2. Analysis of data gathered from observation										
3. Lay-outing, Editing and Printing of 100 copies of coffee table book										

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

Provide concrete database on sorghum industry in the Province of Capiz

(15) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

(16) Targets Beneficiaries (Who the clientele are and what are the expected outcome/effects of the use of the projects output)

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(17) Personnel Requirement

Name	Role (Program, Project, Study Leader)	Field of Specialization

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

(18) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount (PhP)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)		
B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)		
- Transportation allowance		
- Meal allowance during data gathering		
- Professional fees (laboratory analysis and other related fees)		
- Miscellaneous (for incidental expenses)		
- Printing		
C. CAPITAL OUTLAY		
Total		

(19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

References:

Morais Cardoso L. Pine SS, HS Martino, Pine-Ana HM. 2015. Sorghum: Nutrient, Bioactive Compounds, and Potential Impact on Human Health. Rev. Food Science Nutrient

Belum V.S. Reddy, A Ashok Kumar and P. Sanjaing Reddy 2010. Recent Advances in Sorghum Improvement Research at ICRISAT. Kasetart J (Nat Sci) 44: 499-506.

Yi-Hong Wang and Ismail Dweikat, 2016. Genetic and Genomic Resources for Grain Cereals Improvement

<https://www.purdue.edu › 0004.Hamaker.sorghum.htm>

<https://www.healthline.com/nutrition/sorghum#bottom-line>

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(20) Capsule Curriculum Vitae (One-page brief C.V. only)